

CLINICAL TRIAL DATASETS AND THE IMPORTANCE OF DATA QUALITY IN CLINICAL TRIALS

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Abstract

High-quality clinical trial data directly impacts patient safety outcomes and the approval of clinical findings by regulatory agencies. It emphasizes the shift from traditional data cleaning to active, AI-driven data cleaning and standardization in accordance with Clinical Data Interchange Standards Consortium (CDISC) Study Data Tabulation Model (SDTM), fitting clinical trial data “for purpose.” The paper addresses the ways in which machine learning technology can be incorporated with data safety checkpoints to guarantee data integrity, traceability, and analytical accuracy in complex clinical investigations compliant with CDISC’s SDTM.

Introduction

Clinical trial datasets comprise a substantial amount of information on patient demographics, medical history, laboratory investigations, adverse events, and treatment outcomes. The interpretation of therapeutic efficacy and safety is exclusively based on the statistical analysis of the indicated data, which must be comprehensive and correct to ensure trial conclusion validity.

Thus, missing, repeated, and inconsistent data items negatively intervene with the accuracy of evidence generated by clinical trials, adversely impacting regulatory decision-making and patient safety outcomes. Clinical data management (CDM) ensures the implementation of adequate measures to plan, control, and prepare datasets for statistical analysis. In addition, knowledge of the characteristics of clinical trial datasets is paramount to developing effective AI-powered data cleaning and standardization approaches.

Clinical Trial Datasets and Their Role

This section highlights the basic types of clinical trial datasets utilized in statistical analysis. For clarity, the data items were separated into categories. These datasets enable statistical comparison of treatment groups in terms of primary and secondary endpoints, supporting the regulator’s decision on trial product approval. This section also elaborates on the criticality of data quality, including the importance of avoiding data biases that impair analysis validity.

Importance of Data Quality in Clinical Trials

Data quality in clinical trials is defined by a set of criteria that determine the degree to which study data are complete, accurate, consistent, and sufficiently reliable to provide credible interpretations and support appropriate decision-making. In contrast, poor data quality is

associated with the occurrence of distorted trial results, erroneous conclusions, and ineffective regulatory decisions on investigational product approval.

Indeed, clinical trial data integrity, accuracy, and consistency are considered by regulatory agencies as a pivotal element of study validity and reliability. In addition, it is noteworthy that in an AI-powered framework, data flaws can be automatically detected and flagged; however, the structure of such data must be specifically designed.

Missing Data

Missing data present in clinical trial datasets account for a significant source of bias that undermines the accuracy of statistical analysis and, consequently, the validity of trial conclusions.

Missingness occurs when particular data items or records are inadequately captured or registered, rendering them useless for statistical analysis. The absence of data items can be caused by patient non-attendance, clinical site deficiencies, data entry errors, or technical malfunctions. Notably, missing data can be informative, particularly when associated with the drop-out mechanism, in which patients discontinue study treatment, which biases the estimate of an otherwise positive treatment effect. Thus, CDM practices and biostatistical approaches are paramount in mitigating the adverse impact of missing data items on trial results.

Prior to the commencement of a clinical trial, the CDM team establishes a list of critical data items and patient visits to help design an efficient data collection strategy and minimize the occurrence of missing data. Concurrently, during the conduct of a trial, central monitoring allows the CDM team to evaluate the rate of data completeness and response, informing the development of corrective measures if the extent of missing data exceeds the predefined threshold. The data management plan (DMP) enables the CDM team to identify the most appropriate statistical approach to addressing missing data items, ranging from database exclusion to application of mixed-model analysis. In addition, the CDM team can utilize sensitivity analysis to evaluate the effect of different dropout mechanisms on trial results. An illustrative example of the impact of missing data on statistical analysis is provided by the student dataset.

In addition to missing data items, the occurrence of duplicate records and inconsistencies also affects the integrity of clinical trial data. Duplicate records and inconsistencies present another source of data imperfection that must be eliminated during the data cleaning process. In particular, record duplication is a common occurrence in clinical trials due to the frequent number of patient interactions throughout the trial conduct. Thus, the occurrence of duplicate records can severely impact trial results, particularly when the presence of such records affects treatment-related adverse events.

Duplicates can be removed upon the results of consistency checks conducted by the CDM team. In addition, the CDM team can benefit from utilizing statistical software that can be programmed to alert researchers to inconsistent data items requiring manual verification and record elimination. For instance, duplicated rows of a patient dataset can be easily removed using statistical software.

Similarly, logical inconsistencies, such as the presence of conflicting date formats, also impact the accuracy of data analysis. One of the common examples of logical inconsistencies is the occurrence of impossible values, which stem from differences in data collection practices or variations in case report form design across different clinical sites.

Inconsistencies can be resolved during the data cleaning process, during which the CDM team examines the presence of duplicated records, inconsistent field labels, and values that do not correspond to the established field formats. In addition, the CDM team can utilize automated procedures that can help identify duplicated records, values that do not conform to the set field formats, and other sources of data inconsistencies. For instance, finding duplicates and fields with inconsistent format can be demonstrated using a sample patient dataset.

The Importance of Standardization

Data standardization is aimed at ensuring that data across different sites and trials are fit for the purpose, enabling the transformation of datasets into a uniform format required for analysis. Data standardization in clinical trials entails the use of common data elements and terminology as well as standardized case report forms and common data operation models developed in accordance with global data formats, including Clinical Data Interchange Standards Consortium (CDISC) and Study Data Tabulation Model (SDTM). Standardization supports the quality of clinical trial data, facilitates audit trail inspection, and promotes evidence synthesis for regulatory decision-making.

Conclusion

This paper provides an overview of the key principles of clinical trial data quality, ranging from the impact of missing data and record duplication on statistical analysis to the role of data standardization in enhancing data integrity and regulatory compliance.